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| **Fiscal Year:** | FY 2015 | **Task Last Updated:** | FY 12/30/2014 |
| **PI Name:** | Wiese, Claudia Ph.D. | | |
| **Project Title:** | HZE-induced skin carcinoma: an in-vivo-like multicellular epithelial model system for analysis of radiation quality effects and DSB repair processes | | |
| **Division Name:** | Human Research | | |
| **Program/Discipline:** | HUMAN RESEARCH | | |
| **Program/Discipline--Element/Subdiscipline:** | HUMAN RESEARCH--Radiation health | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | (1) **SR**:Space Radiation | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **Comments:** | For information purposes only--PI moved in June 2014; however, funding did not go with her. As of 6/1/14, Colorado State University, Department of Environmental and Radiological Health Sciences, 485 MRB - 1618 Campus Delivery, Fort Collins, CO 80523-1618, Office: (970) 491 7618, Email: Claudia.Wiese@colostate.edu | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | 2010 Space Radiobiology NNJ10ZSA001N |
| **Start Date:** | 04/01/2011 | **End Date:** | 10/01/2014 |
| **No. of Post Docs:** | 0 | **No. of PhD Degrees:** | 0 |
| **No. of PhD Candidates:** | 0 | **No. of Master' Degrees:** | 0 |
| **No. of Master's Candidates:** | 0 | **No. of Bachelor's Degrees:** | 1 |
| **No. of Bachelor's Candidates:** | 1 | **Monitoring Center:** | NASA JSC |
| **Contact Monitor:** | Simonsen, Lisa | **Contact Phone:** | |
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| **Flight Program:** | | | |
| **Flight Assignment:** | NOTE: End date changed to 10/1/2014 for final reporting purposes (Ed., 1/5/2015)<br>NOTE: Gap changes per IRP Rev E (Ed., 3/25/14)<br>NOTE: Extended to 9/30/2014 per PI (Ed., 1/28/14) | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | Snijders, Antoine ( Lawrence Berkeley National Laboratory ) | | |
| **Grant/Contract No.:** | NNJ11HB91I | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

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| **Task Description:** | There is considerable evidence that ionizing radiation causes non-melanoma skin cancer (NMSC), and the risk for NMSC in astronauts is increased. To further assess the cancer risk for humans from space radiation, an in-vivo-like multicellular skin model system is used. Multiple different endpoints as surrogate markers for skin cancer are assessed. Persistent chromosomal damage, altered cell proliferation, and evasion from apoptosis are studied in 2D monolayer cultures of normal primary and of initiated human keratinocytes exposed to a variety of charged nuclei (Aim 1). In addition, to more realistically recapitulate the complex architecture of a tissue, an in-vivo-like skin model system, composed of fibroblasts and keratinocytes, is used to ascertain the risk of space radiation-induced cancer development processes in 3D (i.e. persistent chromosomal damage, altered cell proliferation, evasion from apoptosis, and differentiation) (Aim 2). Last, a surface transplantation assay of exposed 3D reconstructed human epithelia onto immunodeficient mice will be carried out in future to test for intermediate and late events in cancer progression (cell invasion and metastasis, respectively) (Aim 3). We hypothesize that, in basal cells, unfaithful DNA DSB repair will lead to persistent chromosomal damage, promoting altered cell growth and escape from tissue homeostasis. We believe that the proposed approach is highly relevant for assessing the risk of skin cancer in astronauts and for modeling space radiation-induced tumor development in other epithelial tissues. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | Skin as a model system to study radiation-induced human epithelial cancers: Skin is one of the most complex organs of the body, consisting of multiple cell types, including epithelial keratinocytes and connective tissue (i.e. fibroblasts). Keratinocytes form a stratified squamous epithelium and are composed of one basal cell layer and several differentiated cell layers. Proliferation is limited to the basal layer, whereas differentiating cells withdraw from the cell cycle and express various differentiation markers, dependent on the stage of their differentiation. Because radiation-induced neoplasms in skin derive primarily from epithelial cells, radiobiological studies on keratinocyte monolayers (2D), as suggested by us, provide a good first approximation for assessing cancer development processes triggered by ionizing radiation, but also by other environmental mutagens. Although 2D in vitro assays are high throughput and can be very informative, they do not appropriately recapitulate the complex interactions between different cell types and the extracellular matrix within tissues. The in vivo animal model systems may provide the complex native environment, but are not well suited for experimental manipulation and may not accurately recapitulate the human condition.<br>Three-dimensional (3D) in vitro organotypic culture models of epithelial tissues, like the one developed by us, provide an intermediate solution by recapitulating the major features of normal 3D tissue organization using human derived cells, and hold great potential for studying the effects of radiation and other environmental mutagens on tissue homeostasis. In this model system, keratinocytes are grown on a collagen layer containing fibroblasts and are exposed to air. These conditions allow fibroblasts and keratinocytes to interact such that a stratified and differentiated epithelium is formed, resembling the architecture of the human skin. For the experiments proposed in our study, we used low-passage normal primary human keratinocytes (NPKs). The advantage of primary keratinocytes is that their response most closely resembles that of normal skin. To mimic intraepithelial neoplastic lesions, we also assessed immortalized non-tumorigenic HaCaT cells. HaCaT cells were derived from normal skin of a male patient and spontaneously immortalized in culture. However, HaCaT cells retain some functional differentiation properties of normal keratinocytes, form a somewhat stratified squamous epithelium when grown in organotypic cultures (despite the absence of fully functional p53), and are non-tumorigenic when injected into mice. Notably, HaCaT cells carry a UVB fingerprint mutation in p53 that resides in many skin cells of healthy middle-aged individuals. This model for intraepithelial neoplasia is highly relevant since it has been shown previously that adult skin contains clonal patches of p53 mutated keratinocytes approximately 60-3000 cells in size, which can involve up to 4% of the epidermis. We believe that our studies have relevance for a better understanding of the processes involved in the development of epithelial cancers due to exposure to environmental mutagens. For example, the skin model system, consisting of co-cultured fibroblasts and keratinocytes is one of the best-understood human tissue systems, and fibroblast-derived soluble factors play an important role in maintaining the well-controlled balance between epithelial proliferation and differentiation. For human skin, many of these paracrine factors and the pathways they are involved in are known, and they also are conserved across other epithelial human organs. Therefore, the response of skin tissue to environmental factors is relevant for predicting cancer risk in other tissues of epithelial origin.<br>Taken together, our results show that the fraction of DNA damage remaining (as determined by micronucleus analysis) after high LET heavy ion exposure is dependent on tissue architecture and the state of cellular immortalization. We suggest that these factors should be considered when estimating the risk of radiogenic skin cancer and other epithelial cancers in astronauts on deep space missions. |
| **Task Progress:** | The cytokinesis-block micronucleus (MN) assay was used to assess the genotoxicity of low doses of different types of space radiation. Normal human primary keratinocytes and immortalized keratinocytes grown in 2D monolayers each were exposed to graded doses of 0.3 or 1.0 GeV/n silicon ions or similar energies of iron ions. The frequencies of induced MN were determined and compared to gamma-ray data. RBEmax values ranged from 2.0 to 3.9 for primary keratinocytes and from 2.4 to 6.3 for immortalized keratinocytes. At low radiation doses = 0.4 Gy, 0.3 GeV/n iron ions were the most effective at inducing MN in normal keratinocytes. An “over-kill effect” was observed for 0.3 GeV/n iron ions at higher doses, wherein 1.0 GeV/n iron ions were most efficient in inducing MN. In immortalized keratinocytes, 0.3 GeV/n iron ions produced MN with greater frequency than 1.0 GeV/n iron ions, except at the highest dose tested. MN formation was higher in immortalized keratinocytes than in normal keratinocytes for all doses and radiation qualities investigated.<br>MN induction was also assessed in human keratinocytes cultured in 3D to simulate the complex architecture of human skin. RBE values for MN formation in 3D were reduced for normal keratinocytes exposed to iron ions, but were elevated for immortalized keratinocytes. Overall, MN induction was significantly lower in keratinocytes cultured in 3D than in 2D. Together, the results suggest that tissue architecture and immortalization status modulate the genotoxic response to space radiation, perhaps via alterations in DNA repair fidelity.<br><br>We also have established a surface transplantation assay, which allowed us to extend our investigations of the biological responses of human skin equivalents up to several months post radiation exposure. To do so, male NOD.SCID mice (obtained from Jackson Laboratory) were subjected to the skin grafting procedure at ~ 8 weeks of age and sent to BNL 4 weeks post surgery. Cohorts of 6 mice each were exposed to 1 Gy of 300 or 1000 MeV/n Fe ions, or to 1 Gy of 137Cs gamma-rays for comparison purposes. Six mice were sham irradiated. All mice remained at BNL until necropsy at 8 weeks post radiation exposure. Tissue sections of the human skin grafts and of the undisturbed mouse skin were fixed in 10% buffered formaldehyde for 48 hours, followed by transfer and storage in 70% ethanol until processed and |

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| | embedded in paraffin. Sections of 3 µm each were assessed by H&E staining for tissue architecture, by FISH using mouse and human COT1 DNA for tissue origin, and by IHC for markers of radiation exposure (53BP1 (i.e. genomic instability), Ki-67 (i.e. cell proliferation), and involucrin (i.e. keratinocyte differentiation).<br><br>Our preliminary results show that 53BP1 foci are elevated and persistent in basal keratinocytes of human skin equivalents after radiation exposure. Keratinocytes with locally accumulated nuclear staining for 53BP1 are negative for Ki-67, indicative of a reinforced irreversible growth arrest of cells with persistent genomic instability. We speculate that cellular senescence is accelerated in human basal cells following high LET radiation exposure to limit the proliferation of damaged cells that are at risk for neoplastic transformation, similar to the observations reported by others using low LET IR. |
| **Bibliography Type:** | Description: (Last Updated: 04/11/2018) |
| **Abstracts for Journals and Proceedings** | Snijders AM, Maranon DG, Mannion BM, Leung SG, Moon SC, Kronenberg A, Wiese C. "Micronucleus formation in human keratinocytes is dependent on radiation quality and tissue architecture." High LET radiation effects. Poster presentation at 60th Annual Meeting of the Radiation Research Society, Las Vegas, Nevada, September 21-24, 2014. 60th Annual Meeting of the Radiation Research Society, Las Vegas, Nevada, September 21-24, 2014. , Sep-2014 |
| **Articles in Peer-reviewed Journals** | Snijders AM, Mannion BJ, Leung SG, Moon SC, Kronenberg A, Wiese C. "Micronucleus formation in human keratinocytes is dependent on radiation quality and tissue architecture." Environ Mol Mutagen. 2014 Jul 10. [Epub ahead of print] http://dx.doi.org/10.1002/em.21887 ; PubMed PMID: 25041929 , Jul-2014 |
| **Articles in Peer-reviewed Journals** | Sridharan DM, Asaithamby A, Bailey SM, Costes SV, Doetsch PW, Dynan WS, Kronenberg A, Rithidech KN, Saha J, Snijders AM, Werner E, Wiese C, Cucinotta FA, Pluth JM. "Understanding cancer development processes due to HZE particle exposure: Roles of ROS (reactive oxygen species), DNA damage repair and inflammation." Radiation Research. 2015 Jan;183(1):1-26. http://dx.doi.org/10.1667/RR13804.1 ; PubMed PMID: 25564719 (originally reported as in press, October 2014) , Jan-2015 |